MSDS 570: Trend in global life expectancy over a decade taking socioeconomic and political factors into play.

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1 Introduction

The following proposal will delve into the core logic and framework of our proposed project. Topics to be discussed include the purpose, motivation and reason for choosing our topic, data description and limitations, planned visualization, tools to be explored for our visualization technique, evaluation plan and tentative propose timeline to achieve our objective.

1.1 Purpose

Our group aim to analyze the trend and increase in life expectancy across the world pre-Covid from 2001-2019 based on socioeconomic factors such as country investments in healthcare, education, environmental management and other political factors such as corruption

1.2 Motivation

1.2.1 Our group aim to explore public health and the role of divergent factors such as education, income, social factors such as unemployment and political factors such as Education expenditure

1.2.2 The intended audience includes everyone from policy makers who make policy on budget allocation, students (the role of education in life expectancy), public who vote for legislators . . . etc.)

1.2.3 The decision, insight or story the intended audience should take away is to understand the role and interconnection on how multiple different domain factors affects health care and longevity.

1.3 Research Question

"Is there an influence on longevity based on various socioeconmoic and political factors across global region among all demographics ?"

2 Data Description

Our data source is derived from Kaggle from the world bank

- Visualizations techniques include various types of plots for both categorical and numerical variable.
- We plan to build a series of interconnected visualization to analyze trend in life expectancy based on geolocation, income, education and much more. We also plan to do some more fine-tuning across each column to also see hidden factors such as education level and income level and how that correlates to life expectancy among the same group.

- We then can speculate on whether if there are similarities in life expectancy among each class in groups in the same country or neighboring region irrespective of other factors such as education level, unemployment or communicable / non-communicable diseases. A base comparison will be offered across each continental region.

- Finally, we can form a conclusion on which variables are more predictive in life expectancy and if other factors such as investment in education, health care expenditure, sanitation are crucial in each region/country which paints a more accurate picture of disparity among each region rather than a broad overview of life expectancy.

- Benefit of this includes trying to create or target policy that helps improve life expectancy globally. Evaluation will be based on peer feedback, task-based test and heuristic checklist

2.1 Data Information

The name of the dataset is life expectancy and socio-economic (world bank)

- Source: Kaggle

- Link is:`https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank?utm_source=chatgpt.com`

- Analysis of dataset: rows are divided into (Country Name, Country Code, Region, Income Group, Year, Life Expectancy World Bank, Prevalence of Undernourishment, CO2, Health Expenditure %, Education Expenditure, Unemployment, Corruption, Sanitation, Injuries, Communicable, Non-communicable)

2.1.1 Data Size: Variables we will use

- No of rows are 3307

- No of columns are 16

- range: 2001 – 2019

2.1.2 Known Limitations

- Country Name: Unique variable is 174

- Country Code: Unique variable is 174

- Region: Unique variable is 7

- Income Group: Unique variable is 7

- Year: Missing 0, Mismatched 0

- Life Expectancy World Bank: Missing 188 variable

- Prevalence of Undernourishment: Missing 684

- CO2: Missing 152

- Health Expenditure %: Missing 180

- Education Expenditure: Missing 1090

- Unemployment: Missing 304

- Corruption: Missing 2331

- Sanitation: Missing 1247

- Injuries: Missing 0
- Communicable: Missing 0
- Non-communicable: Missing 0

3 Planned Visualization

Visualization includes bar chart, line chart, scatter plot, bubble charts. The aim of the project is to visualize and see trend in life expectancy over decades by looking at subtle factors that are determinate on life expectancy. The question the visualization answer is based on if health and wellness trend positive according to time or if there are other mitigating factors that influence health and wellness. Visual encoding to be explored includes warm and contrasting color to visualize data that span decades, size,

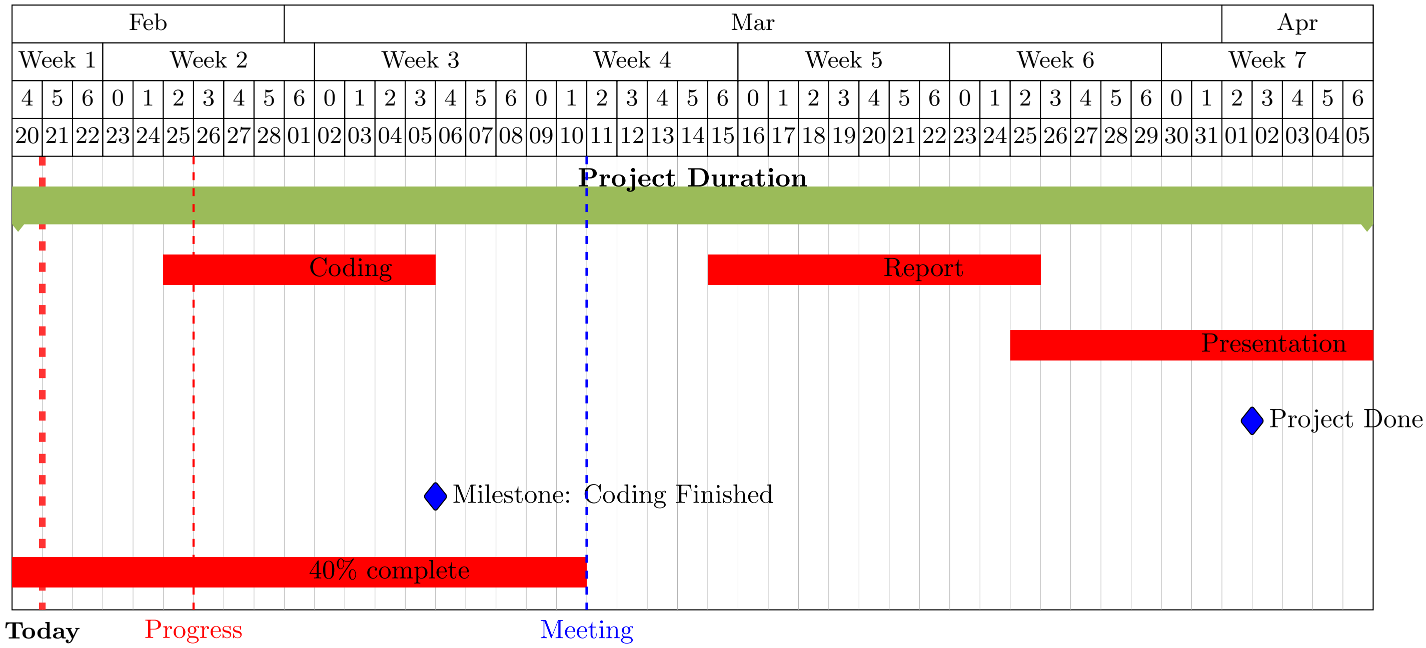
4 Tools & Workflow

- Tool and workflow will primarily be done on Python (using the pandas package and matplotlib).
- Coding resources will either be done on Jupyter notebook or google Colab
- The workflow plan will have a visual flow chart will includes cleaning the data (checking for nan), imputation (filling nans), data exploration, design iterations and final story
- Version control includes the creation of a repository named MSDS 570 and commits which will be snapshots of coding progress

5 Evaluation Plan

- Task-based test (aim to predict trend in life health trend visually to answer basic question such as influence of education attainment in health, influence of income or class, influence of sanitation in health)
- Peer feedback: We encourage our peers to cross-examine the project visualization and critically analyze subtle information that can lead to the trend observed in the data.
- Heuristic checklist: color contrast (since some of the categorical data in each column is broken down to different groups, using color and contrast to highlight changes across column become important).

6 Timeline



7 Expected Deliverables

- Cleaned datasets: achieved via data imputation
- Final visualization: Visualization story
- Short write-up (summary of each visual plot and depiction of trends across years, final report project)
- Code/notebook: Google colab using pandas to fill missing NaN, running data inspection to check for number or row, columns, getting the info of the data such as the mean, range, and median (which could be used for imputation to fill NaN in missing values), etc.
- References: Citation from World Bank.